

Т₂

с) оценка по ЧПТ:

$$j \sim \text{Exp}(1)$$

$$\frac{\sum j_i - n M_j}{\sqrt{n D_j}} \rightsquigarrow N(0, 1)$$

$$M_j = \frac{1}{\lambda} = 1$$

$$D_j = \frac{1}{\lambda^2} = 1$$

$$\frac{\bar{X} - M_j}{\sqrt{D_j}} \rightsquigarrow N(0, 1)$$

$$\frac{\bar{X} - 1}{\sqrt{25}} \rightsquigarrow N(0, 1)$$

$$\bar{X} \rightsquigarrow N(1, \frac{1}{25})$$

е) медiana 25 элементов $\rightarrow$ 13-й элемент, т.е. $X_{(13)}$

$$j_{(13)} \sim \sum_{i=13}^{25} C_{25}^i F^i(x) (1-F(x))^{25-i} = \varphi(x) - \text{функция распред.}$$

$$\varphi(x) = \varphi(x)' = \sum_{i=13}^{25} \left[C_{25}^i \left(i F(x)^{i-1} \cdot \underbrace{F'(x)}_{p(x)} (1-F(x))^{25-i} - (25-i) F^i(x) (1-F(x))^{24-i} \cdot \underbrace{F'(x)}_{p(x)} \right) \right] \quad \text{е)}$$

$$p(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & x < 0 \end{cases}$$

$$\left\{ \begin{array}{l} x \geq 0: F(x) = \int_0^x e^{-t} dt = -e^{-t} \Big|_0^x = 1 - e^{-x} \\ x < 0: F(x) = 0 \end{array} \right.$$

$$\text{е)} \sum_{i=13}^{25} \left[C_{25}^i \left(i (1-e^{-x})^{i-1} \cdot e^{-x} (e^{-x})^{25-i} - (25-i) (1-e^{-x})^i (e^{-x})^{24-i} \cdot e^{-x} \right) \right]$$